Early Detection Through Comprehensive Screening Approach For Children To Distinguish Intellectual Disablity From Effort Variablity

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***Abstract*—Early identification of intellectual disability (ID) in children aged 0–9 years remains a persistent challenge, particularly in distinguishing genuine cognitive impairment from low performance arising due to fluctuating effort, attention, environmental exposure, or cultural and language factors [3], [7]. Conventional assessment practices, including standardized intelligence tests and static observational methods, typically capture only short, context-dependent snapshots of behavior, which can lead to misclassification, delayed intervention, and inappropriate educational placement [7]. To overcome these limitations, recent advances increasingly explore technology-enabled, longitudinal assessment paradigms supported by artificial intelligence (AI) [4]. This work presents a behavioral game-based screening framework that leverages repeated, child-friendly interactive micro-games to capture fine-grained behavioral telemetry—such as reaction time, accuracy, hesitation patterns, motor precision, and engagement consistency—across multiple sessions [1], [2]. Unlike existing digital screening tools that rely primarily on aggregated performance scores, the proposed system explicitly models temporal stability and variability using a hybrid AI/ML pipeline combining interpretable tree-based models with temporal deep learning architectures. A novel Child Effort–Cognition Index (CECI) is introduced to integrate performance consistency, volatility, and inferred effort into a single transparent score that supports early risk stratification while preserving clinical interpretability.By shifting early screening from one-time testing to continuous behavioral monitoring, the proposed approach aims to reduce false positives driven by situational factors and improve early referral decisions, particularly in low-resource settings where specialist access is limited [1], [4]. The framework is positioned as a decision-support tool rather than a diagnostic replacement, with the potential to enhance early intervention planning, promote inclusive education, and improve long-term developmental outcomes for children at risk of intellectual disability.**

***Keywords*—Intellectual disability, early childhood screening, gamified assessment, behavioral analytics, machine learning, explainable AI, inclusive education.**

I. Introduction

Accurate identification of intellectual disability (ID) during early childhood remains a complex and unresolved challenge in developmental psychology, special education, and clinical practice [7]. Children between 0 and 9 years of age exhibit wide variability in cognitive maturation, behavioral regulation, and learning pace, influenced by socio-cultural, linguistic, and environmental factors. This natural heterogeneity complicates the task of distinguishing genuine neurodevelopmental impairment from performance limitations caused by transient factors such as low motivation, attentional fluctuation, limited prior exposure, or situational stress [3], [7]. Conventional assessment methodologies rely primarily on standardized intelligence tests and brief observational evaluations conducted in controlled settings. While these instruments are clinically established, they provide only short, context-dependent snapshots of behavior and are sensitive to non-cognitive influences including fatigue, anxiety, and examiner interaction [7].

As a result, static assessments often fail to capture the dynamic and effort-dependent nature of early childhood performance, leading to uncertainty in whether observed developmental delays reflect stable cognitive limitations or contextual variability [3]. This ambiguity contributes to both delayed intervention for children with genuine needs and inappropriate placement of typically developing children into special education programs, with long-term educational and psychosocial consequences [3]. Recent advances in digital assessment technologies and artificial intelligence (AI) offer opportunities to overcome these limitations by enabling longi-

tudinal, behavior-centered screening paradigms [4]. Gamified and interactive environments allow children to be assessed repeatedly in engaging, low-anxiety contexts, generating rich behavioral telemetry such as response timing, accuracy patterns, hesitation, and engagement consistency across sessions [1], [2]. However, many existing digital tools still aggregate performance into single difficulty scores and do not explicitly model how effort and engagement fluctuate over time.

Motivated by this gap, this work proposes a game-based behavioral screening framework that emphasizes temporal consistency, effort variability, and interpretability as core design principles. The system integrates repeated cognitive micro-games with a hybrid machine learning pipeline and introduces a transparent Child Effort–Cognition Index (CECI) to differentiate persistent cognitive difficulty from inconsistent effort. Positioned as a decision-support tool rather than a diagnostic replacement, the proposed approach aims to improve early risk stratification and referral decisions, particularly in low-resource settings where access to specialist assessment remains limited.

II. Importance and Significance of the Topic

Timely identification of intellectual disability (ID) during early childhood is strongly associated with improved developmental, educational, and social outcomes, as early intervention is most effective during periods of high neural plasticity [1], [4]. Inaccurate or delayed screening, however, can result in inappropriate educational placement, inefficient use of support resources, and long-term psychosocial consequences for children and their families [3], [7]. From a systems perspective, misclassification places a sustained burden on educational and healthcare infrastructures, particularly in public and low-resource settings where specialist assessment capacity is limited. Consequently, improving the precision and scalability of early screening has become a key priority within inclusive education and early intervention frameworks worldwide [4].

Technology-enabled assessment platforms offer an opportunity to address these challenges by moving beyond episodic testing toward continuous developmental observation. Interactive digital systems can support repeated, low-cost screening while capturing behavioral signals that are difficult to observe reliably in single-session clinical assessments [1], [2]. When combined with data-driven analytics, such approaches have the potential to support earlier, more informed referral decisions and to extend access to effective screening in environments where traditional diagnostic resources are scarce.

III. Background and Target Audience

This research is targeted at a multi-disciplinary community of researchers, practitioners, educators, policymakers, and technologists working in the area of early childhood development, special education, and AI-driven healthcare solutions [4]. The early detection of developmental issues demands a multi-disciplinary approach, as cognitive, behavioral, and educational variables interact with each other in determining the outcome of screening and intervention procedures. The proposed framework integrates traditional psychological evaluation procedures with AI-driven behavioral observation tools to enable more objective and scalable assessment of child development [2], [4].

For practitioners and therapists, the system offers longitudinal behavioral data based on repeated digital engagement, facilitating informed referral and intervention planning beyond single-session screening [3]. Educators and special education professionals can leverage these indicators to detect children who may need early academic support.From a policy point of view, the proposed framework is well-aligned with the initiatives of inclusive education, as it enables scalable screening in low-resource environments where access to specialists is not feasible [4]. From the perspective of technology developers and AI researchers, this study exemplifies how digital assessment, behavioral analysis, and machine learning can be combined in child-friendly applications that emphasize interpretability, usability, and ethics [1], [2]. This work serves as a reference point for all stakeholders interested in efficient, scalable, and interpretable screening solutions that can complement the conventional assessment process.

IV. Surveyed Research Area and Its Justification

The surveyed research area—technology-augmented assessment of developmental and intellectual disabilities—has received increasing attention due to its potential to transform early childhood screening practices [1], [4]. Prior studies consistently highlight the limitations of conventional assessment approaches that rely heavily on standardized, static tests and brief observational measures, which often fail to capture contextual and temporal variability in child performance [3], [7]. In response, recent research has explored alternative methods based on gamified interaction [2], [6], behavioral analytics [1], and AI-driven monitoring frameworks [4]. These approaches enable continuous data capture, improved engagement, and fine-grained behavioral analysis, offering richer developmental insights than single-session evaluations. However, much of the existing work focuses on aggregate performance outcomes and does not explicitly address the challenge of distinguishing persistent cognitive difficulty from effort-related or situational variability.

This research area is therefore critical not only for surveying existing digital and AI-enabled assessment frameworks, but also for identifying gaps where interactive, longitudinal, and explainable systems can better support early identification and targeted intervention. Fig. 1 summarizes the surveyed research domains and motivates the need for integrated, effort-aware, AI-driven screening platforms tailored to early childhood settings.

V. Techniques and Classification

Accurate early screening of intellectual disability in young children requires analysis of behavioral patterns that evolve over time rather than reliance on single-session performance.

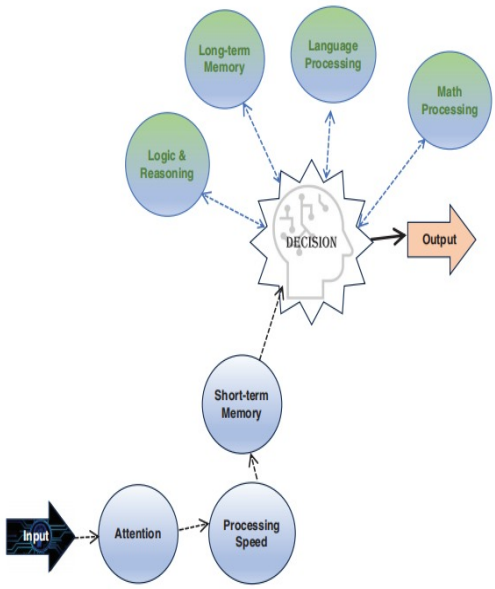

Fig. 1. Surveyed research domains in technology-augmented developmental assessment.

Conventional assessments often provide only limited snapshots of behavior and may fail to distinguish persistent cognitive impairment from effort-related variability [3], [7]. To address these limitations, the proposed framework adopts a hybrid machine learning approach that captures longitudinal effort patterns while maintaining interpretability and calibrated risk estimation. The system begins with behavioral telemetry collected from repeated interactions with gamified cognitive tasks across multiple sessions. This multi-session data includes reaction time, accuracy, hesitation patterns, and engagement indicators. Digital game-based assessments enable continuous observation in low-anxiety environments and have been shown to produce richer behavioral signals than traditional testing methods [1], [2]. Variability across sessions is treated as diagnostically meaningful, as fluctuations in performance may reflect inconsistent effort rather than stable cognitive difficulty. To analyze this data, three complementary modeling components are employed. First, tree-based models such as Random Forest or XGBoost operate on aggregated tabular features to generate an initial risk estimate. These models perform well with structured behavioral data and provide interpretable feature importance measures, allowing identification of key indicators influencing performance [4]. However, they lack temporal awareness and cannot capture changes across sessions.

Second, temporal deep learning models, including Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) networks, process ordered session sequences to learn performance trends over time. Such models capture patterns of improvement, stability, volatility, or decline, enabling differentiation between persistent cognitive difficulty and inconsistent effort. Temporal analysis is particularly important in early childhood, where developmental trajectories may vary widely [4]. Third, a Bayesian logistic model serves as a calibration layer that produces probabilistic outputs with associated uncertainty estimates. This component is especially useful when data is sparse, risk levels are borderline, or model predictions conflict. Incorporating uncertainty helps prevent overconfident decisions in sensitive screening contexts and supports safer referral recommendations.

The outputs of these models are integrated into a single interpretable metric termed the Child Effort–Cognition Index (CECI). This index combines evidence from multiple analytical perspectives to produce a stable decision cue while remaining transparent to clinicians and caregivers. Risk levels are categorized into bands: green indicating likely effort-related variability, amber indicating the need for monitoring, and red suggesting referral for specialist evaluation. Importantly, the framework functions as a decision-support system rather than a diagnostic tool. Results are delivered through dashboards tailored to different stakeholders, including simplified indicators for parents and teachers and detailed trend analyses for clinicians. Human oversight remains central, ensuring that automated insights complement professional judgment and support continuous reassessment as additional behavioral data becomes available [4].

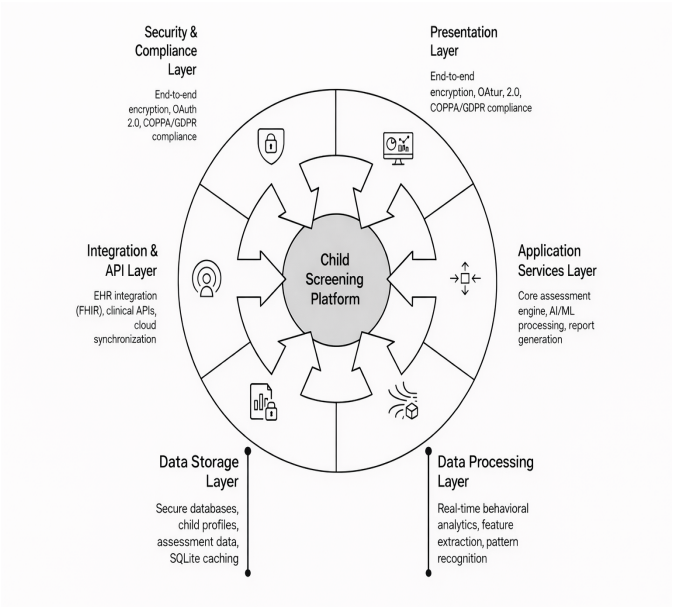

Fig. 2. Illustrates the multi-layer architecture of Child Screening Platform

Above Figure shows the multi-layered architecture of the proposed child screening system. At the center of the architecture is the screening engine, which is supported by layers of functionality that handle various tasks in operating the system. The Data Processing Layer is responsible for real-time analysis of behavior and extraction of features, while the Data Storage Layer is responsible for maintaining secure databases of child profiles and assessment results. The Application Services Layer supports the core assessment engine,

AI/ML processing engines, and report generation tools. The Presentation Layer is responsible for providing interfaces to the system's stakeholders, which can be used for visualizing results and progress. System integration is supported by the Integration and API Layer, which supports clinical APIs and cloud synchronization. Finally, the Security and Compliance Layer is responsible for enforcing security and regulatory requirements, including encryption, authentication, and compliance with standards such as GDPR and COPPA.

VI. Breakthroughs in Assessment Technology

A major breakthrough in early childhood intellectual disability assessment is the shift from episodic, subjective evaluations toward continuous, data-driven screening frameworks. Advances in artificial intelligence (AI) enable assessment systems to adapt dynamically to a child's interactions, capturing fine-grained behavioral patterns—such as response timing variability, hesitation frequency, and engagement stability—that are often overlooked in conventional clinical testing [4]. This transition allows assessment to reflect real-world behavior across time rather than performance within a single, constrained session. Gamified digital assessment environments represent a critical component of this transformation. Empirical studies indicate that game-based tasks significantly improve engagement and task persistence in young children, thereby increasing the validity of observed performance by reducing anxiety, fatigue, and disengagement effects commonly seen in formal testing contexts [2], [6]. By embedding assessment within play, these systems encourage children to demonstrate their functional abilities more naturally, producing richer and more reliable behavioral data.

In parallel, modern digital platforms facilitate improved collaboration among families, educators, and healthcare professionals. Automated reporting, risk alerts, and milestone-based visualization dashboards enable stakeholders to monitor developmental progress continuously rather than relying on infrequent evaluations. These feedback mechanisms support early identification of atypical patterns and enable timely adjustments to intervention strategies. Importantly, such systems promote adaptive care pathways, where screening outcomes evolve alongside the child's development rather than remaining static labels. Collectively, these technological advancements redefine early developmental screening by integrating AI-driven analytics, gamified interaction, and longitudinal monitoring. The resulting systems provide a scalable foundation for effort-aware, interpretable, and context-sensitive assessment, bridging the gap between traditional diagnostic tools and emerging intelligent screening platforms.

The pipeline begins with the collection of behavioral interaction data generated when children engage with gamified cognitive tasks such as memory games, problem-solving activities, and language-based exercises. These interactions produce behavioral telemetry including response time, task accuracy, hesitation patterns, and interaction behavior. The collected data is processed through a telemetry processing stage where

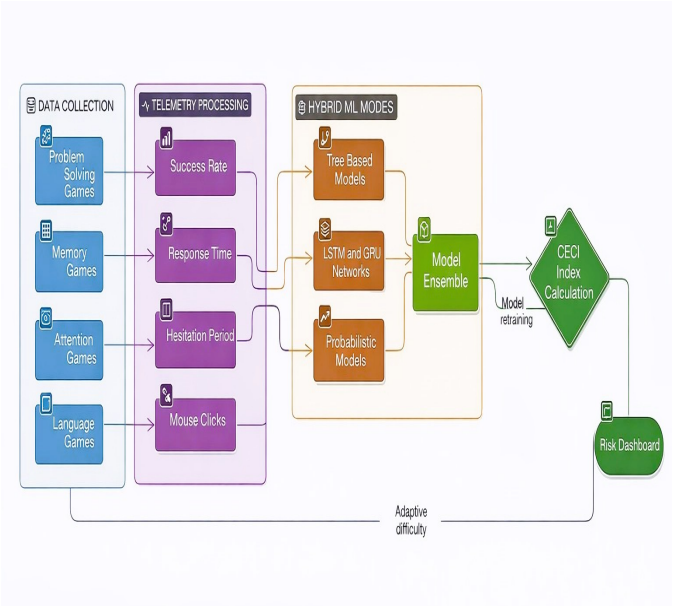

Fig. 3. Illustrates the hybrid machine learning pipeline designed for effort-aware screening of intellectual disability in children.

relevant behavioral features are extracted and structured for analysis. The processed features are then analyzed using a hybrid set of machine learning models, including tree-based models for interpretable feature-based prediction, temporal deep learning models for analyzing performance trends across sessions, and probabilistic models for improving prediction reliability and uncertainty estimation. The outputs of these models are combined through a model ensemble mechanism to generate a unified assessment. Finally, the integrated output is used to compute the Child Effort–Cognition Index (CECI), which summarizes the child's behavioral patterns and provides risk-based screening results through an interactive dashboard for parents, educators, and clinicians. This pipeline enables continuous monitoring and improves the reliability of early developmental screening.

A. AI/ML Modeling and Effort–Cognition Formalization

The proposed screening framework models child performance as a longitudinal behavioral process rather than a single-session outcome. Each child interacts with a set of gamified cognitive micro-games across multiple sessions, generating trial-level interaction data that are aggregated into session-level and longitudinal features for machine learning analysis.

Let $y_{ij} \in \{0, 1\}$ denote the correctness of trial i in session j, where a value of 1 indicates that the child's response matches the expected outcome for the task and 0 indicates an incorrect response. The expected outcome is defined by the predefined correct answer or target behavior associated with each game task. Thus, the difference between expected and actual performance is captured in a binary manner, reflecting whether the child's response satisfies the task requirements.and

let N_j be the total number of trials in that session. The session-level accuracy is computed as:

$$\mathrm{Acc}_j = \frac{1}{N_j} \sum_{i=1}^{N_j} y_{ij} \tag{1}$$

In addition to accuracy, other session-level features such as mean reaction time, hesitation ratio, and task completion rate are derived from raw telemetry and collectively form a feature vector $\mathbf{x}_j$ for each session. Over S sessions, a child is represented as an ordered temporal sequence:

$$\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_S\} \tag{2}$$

To quantify longitudinal stability and effort-related variability, cross-session performance volatility is measured using the variance of session accuracies:

$$\mathrm{Var}(\mathrm{Acc}) = \frac{1}{S} \sum_{j=1}^{S} (\mathrm{Acc}_j - \mu_{\mathrm{Acc}})^2 \tag{3}$$

where μ_{Acc} denotes the mean accuracy across all sessions. High volatility values typically indicate inconsistent engagement or fluctuating effort, whereas persistently low but stable accuracy suggests possible underlying cognitive difficulty.

The machine learning pipeline comprises three complementary components:

(i) tree-based models (e.g., Random Forest or XGBoost) trained on aggregated session and longitudinal features to provide interpretable baseline predictions;
(ii) temporal deep learning models (LSTM/GRU) that process ordered session sequences to estimate persistent cognitive difficulty and effort consistency; and
(iii) a lightweight probabilistic model to improve calibration and uncertainty estimation for high-risk cases.

The outputs of these models are integrated into a Child Effort–Cognition Index (CECI), designed to provide a transparent and clinically interpretable summary score:

$$\mathrm{CECI} = w_1 \cdot P_{\mathrm{ID}} + w_2 \cdot \big(1 - \mathrm{Var}(\mathrm{Acc})\big) - w_3 \cdot P_{\mathrm{Eff}} \tag{4}$$

where P_{ID} represents the estimated probability of persistent cognitive difficulty, P_{Eff} denotes the probability of low or inconsistent effort, and w_1, w_2, w_3 are normalized weights selected to preserve interpretability. Higher CECI values correspond to stable, persistently low performance indicative of cognitive risk, while lower values suggest that poor performance is more likely driven by effort variability or situational factors.

Rather than producing a diagnostic label, CECI is mapped to coarse risk bands (e.g., *green*, *amber*, *red*) and presented through dashboards to support referral decisions and early intervention planning. This formulation ensures that the proposed system remains explainable, effort-aware, and suitable for decision support, aligning AI-driven analysis with clinical interpretability requirements.

VII. Top Research, Products, and Industry Players

Recent advances in early childhood developmental screening have led to the emergence of both commercial and academic solutions that integrate artificial intelligence, behavioral analytics, and interactive digital environments. In the commercial domain, platforms such as Cognoa leverage similar principles seen in gamified dyslexia detection [9] and tablet-based cognitive tools [12] and Akili Interactive exemplify the growing role of AI-assisted screening and gamified digital therapeutics. These systems demonstrate that structured interaction data collected through child-friendly digital interfaces can support clinically relevant insights, including measurable improvements in engagement and attention-related outcomes. Academic and clinical research institutions continue to expand the methodological foundations of technology-assisted developmental assessment. Organizations such as the Children's Hospital of Philadelphia and the Stanford Center for Biomedical Informatics are advancing multimodal screening protocols that combine behavioral observations with physiological and, in some cases, genetic data to improve diagnostic sensitivity. These efforts highlight the value of integrating diverse data sources to enhance early risk identification while maintaining clinical interpretability.

Within the research literature, stealth and gamified assessment approaches have received increasing attention. Gupta et al.'s BENDI framework for Down syndrome detection illustrates how unobtrusive behavioral measurement embedded within interactive tasks can reduce assessment bias and improve ecological validity [5]. Similarly, Rello et al. demonstrated that online gamified tests can effectively predict dyslexia risk by analyzing interaction patterns rather than relying solely on explicit test responses [6]. These studies underscore the adaptability of game-based behavioral analysis across multiple neurodevelopmental conditions. Despite these advances, most existing systems focus on single-condition screening or aggregate performance outcomes and do not explicitly model effort variability or longitudinal consistency. This gap motivates the present work, which builds upon prior industry and research efforts by emphasizing multi-session behavioral analysis, effort-aware modeling, and interpretable indices designed specifically for early intellectual disability screening.

VIII. Unresolved Issues and Future Research Directions

Despite significant progress in technology-assisted developmental screening, several critical challenges remain unresolved. A persistent difficulty lies in reliably distinguishing genuine cognitive impairment from performance deficits driven by fluctuating effort, attention, or contextual factors, particularly in heterogeneous real-world populations [3]. Behavioral variability is developmentally normative in early childhood, making it challenging for current systems to define stable diagnostic thresholds without longitudinal evidence.

Another unresolved concern is algorithmic bias. Many AI-based screening systems are trained on limited or demographically unbalanced datasets, raising questions about fairness, generalizability, and equity across socioeconomic, cultural, and linguistic groups [4]. Without careful dataset design, validation across diverse populations, and transparent modeling practices, such systems risk reinforcing existing disparities in developmental assessment.

Ethical and regulatory challenges are especially pronounced in pediatric applications. The collection and analysis of fine-grained behavioral data from children necessitate strict adherence to privacy and data protection frameworks such as GDPR and COPPA. Robust parental consent mechanisms, secure data storage, role-based access control, and clear limitations on algorithmic decision-making are essential to ensure responsible deployment. Additionally, accessibility remains a practical barrier, as families in rural or low-resource settings may lack consistent internet connectivity, suitable devices, or digital literacy to fully benefit from advanced screening platforms [1]. From a clinical perspective, while visualization dashboards and automated risk indicators provide valuable decision support, they cannot replace expert judgment. Over-reliance on automated outputs without contextual interpretation may lead to inappropriate conclusions. Consequently, future research must prioritize hybrid assessment models that explicitly integrate AI-driven behavioral analysis with clinician oversight, preserving human interpretability and accountability.

Promising research directions include large-scale, multi-site validation of effort-aware indices, such as longitudinal consistency measures, to establish robust normative baselines. Further integration of multimodal signals—including speech patterns, gaze behavior, and physiological indicators—may enhance sensitivity to subtle cognitive and engagement-related differences. Expanding telehealth-enabled screening frameworks could also improve access in underserved regions, while advances in explainable AI and adaptive learning systems hold potential for more personalized, transparent, and ethically aligned early childhood assessment tools.

Figure 4 shows the design of the multi-level feedback loop system for the continuous improvement of the child screening framework. As shown in the figure, the system includes various levels of feedback mechanisms with different time intervals to ensure the adaptive assessment of the system for improvement. Specifically, the system's performance at the real-time level includes the dynamic assessment of the child's performance through the interaction loop with the gameplay's level of difficulty. Meanwhile, the system learning loop continuously updates the machine learning model with the newly acquired behavioral information for the improvement of the algorithms at the daily or weekly level. Additionally, the parent's feedback loop includes the information provided by the parents through the activities completed at home by the children. Moreover, the clinical feedback loop occurs at the weekly or monthly level to review the results of the assessment for the improvement of the screening indicators. Lastly, the quality assurance loop continuously monitors the performance of the system to ensure the accuracy, reliability, and appropriateness of the system for the development screening of the children.

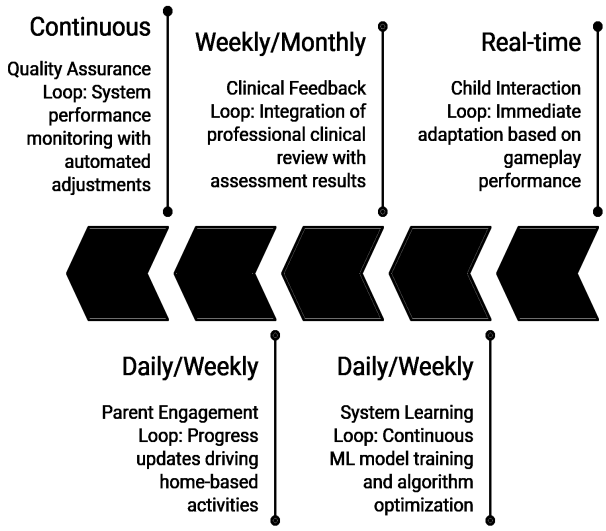

Fig. 4. Multi-Level Feedback Loop System for Continuous Improvement.

IX. Conclusion

Early childhood assessment of intellectual disability remains constrained by the inability of conventional testing methods to reliably separate persistent cognitive impairment from performance variability caused by effort, environmental distraction, or cultural and linguistic factors [3], [7]. Standardized intelligence tests and brief observational checklists offer only isolated snapshots of behavior, which can lead to substantial rates of misclassification and delayed intervention during a developmentally critical period [3], [7]. Such inaccuracies not only affect individual children and families but also place unnecessary strain on educational and healthcare systems. This work addresses these limitations by proposing a technology-assisted, longitudinal screening framework designed for children aged 0–9 years. The proposed system integrates gamified cognitive tasks with continuous behavioral monitoring and milestone-based visualization to capture rich, multi-session interaction data, including response timing, hesitation patterns, engagement consistency, and task completion behavior [1], [2], [4]. By shifting the focus from single-session performance to behavioral trends observed over time, the framework enables a more nuanced understanding of developmental risk.

A key contribution of this approach lies in its emphasis on interpretability and decision support. Rather than replacing formal clinical diagnosis, the system provides caregivers, educators, and therapists with structured observational insights through intuitive dashboards, supporting informed referral decisions and early intervention planning. The proposed design demonstrates how interactive assessment, behavioral analytics,

and AI-enabled monitoring can be combined into a scalable, ethically grounded screening tool that enhances early identification while preserving the role of professional judgment.

X. Further Work

Future work will focus on strengthening the clinical validity, scalability, and generalizability of the proposed screening framework. Large-scale validation studies involving diverse and representative child populations are required to rigorously evaluate the reliability, sensitivity, and specificity of the proposed behavioral indicators and gamified assessment measures in distinguishing persistent intellectual disability from situational performance variability [1], [2], [4]. Close collaboration with educators and clinical psychologists will be essential to align longitudinal behavioral patterns with standardized diagnostic instruments and expert judgments [3], [7]. From a technical perspective, the behavioral analysis pipeline can be extended through multimodal data integration. Incorporating complementary signals such as speech characteristics, gaze and eye-tracking cues, and lightweight physiological indicators may improve estimation of cognitive effort, attention, and engagement when deployed with appropriate consent and privacy safeguards [1], [4]. In parallel, adaptive difficulty adjustment and personalized task sequencing within gamified modules could be explored to better accommodate developmental stage, cultural context, and individual learning profiles [2], [6].

Further improvements will also target system accessibility and deployment feasibility. Enhancing mobile compatibility, multilingual support, and compliance with accessibility standards can increase adoption across socioeconomically and geographically diverse settings [1], [2], [4]. Longitudinal studies leveraging milestone-based dashboards will enable early detection of developmental risk trajectories and support evidence-based intervention planning. Finally, integration with telehealth platforms and secure data exchange mechanisms may facilitate coordinated decision-making among families, therapists, and healthcare providers, while maintaining strict adherence to ethical, privacy, and consent requirements for pediatric data [7]. Collectively, these directions aim to mature the proposed system into a robust, ethically grounded, and scalable early screening tool that complements professional assessment and improves developmental outcomes for children.

References

[1] S. Bhavnani et al., "A non-specialist worker delivered digital assessment of cognitive development (DEEP) in young children: A longitudinal validation study in rural India," PLOS Digital Health, vol. 4, no. 5, Art. no. e0000824, May 2025.

[2] M. Jablonská, I. Fajnerová, and T. Nekovářová, "Cognitive screening for children: Piloting a new battery of interactive games in 4- to 8-year-old children," Cyberpsychology: Journal of Psychosocial Research on Cyberspace, 2024.

[3] J. J. Roor, M. J. V. Peters, B. Dandachi-FitzGerald, and R. W. H. M. Ponds, "Performance validity test failure in the clinical population: A systematic review and meta-analysis of prevalence rates," Neuropsychology Review, 2023.

[4] M. F. Almufareh, S. Tehsin, M. Humayun, and S. Kausa, "Intellectual disability and technology: An artificial intelligence perspective and framework," Journal of Disability Research, 2024.

[5] L. Rello et al., "Predicting risk of dyslexia with an online gamified test," 2016.

[6] National Center for Biotechnology Information, "Intellectual Disability," in Intellectual Disability and Mental Health: A Training Manual in Dual Diagnosis. NCBI Bookshelf, 2015.

[7] A. Alluhaidan, M. Rashid, G. Aldehim, and S. Sakri, "Artificial Intelligence for Neurodevelopmental Disorder Screening: A Systematic Overview," Springer Nature, 2025

[8] E. I. Toki et al., "Machine Learning for Predicting Neurodevelopmental Disorders in Children," Sensors, vol. 24, 2024.

[9] M. S. Singh, N. Mehta, and S. Pandey, "Online Gamified Assessment for Early Detection of Dyslexia Using Machine Learning," Computer Methods and Programs in Biomedicine, 2021.

[10] Jablonska, K. Wrobel, and A. Rogowska, "Cognitive Screening Through Interactive Games: A Comparative Accuracy Study," International Journal of Human–Computer Studies, 2021.

[11] K. McKenzie, D. Paxton, and G. Murray, "Interactive Game-Based Cognitive Screening in Children," Frontiers in Psychology, 2020.

[12] H. Yuan et al., "Evaluation of a Tablet-Based Cognitive Assessment Tool for Children Aged 4–6 Years," PLOS ONE, 2022.